Jump count of the F = 1 target-in-walk arm, single-target ecbench (2026-11-08)

Class. Measured, single-target ecbench sessions under a protocol frozen before data (PROTOCOL.md hashes in `data/frozen.sha256`, freeze commit `a1ce0db2f9`). Counted units only (ecbench GAE, S = cost/sqrt(r)); every comparison at w = 0, 0.1 and 1, under the committed band and under `ecbench band --price-table-work`. Public synthetic Curve25519-family miniatures; no security claim. No pcc claim has been filed; every number is an unverified assertion until it is reviewed [pcc:unverified]. No code change: the existing `jumps` option of the `wwalk` oracle on the `tw=1,cj=1` arm, binary `/home/user/ctarget-f1/release/ecbench` (SHA-256 `d23fab99...` in `data/binary-k{20,24}.sha256`).

Question. The current best single-target arm (`twcj`, 16 jumps) spends 17 % of its k = 20 cost on building jumps. Its jump indices are drawn independently of the point, so the frozen model predicted that 4 or 8 jumps save that set-up with no walk penalty, and that 2 jumps fail.

Answer. The set-up saving is real but the walk penalty is not zero, and the frozen primary prediction failed. With 4 jumps the arm costs 1.220x [1.186, 1.256] the 16-jump arm at k = 20 and 1.268x [1.193, 1.348] at k = 24 (w = 0), because the walk lengthens from 1.39 to 1.86 sqrt(r) and needs 2.93 collisions instead of 2.08 for two independent rows [pcc:unverified]. With 8 jumps the ratio is 1.008 [0.967, 1.051] at k = 20 and 1.050 [1.005, 1.097] at k = 24 [pcc:unverified]. With 32 jumps it is 1.051 [0.996, 1.110] and 1.014 [0.978, 1.051]: the walk is 2.5 % and 1.5 % shorter, the set-up 54 and 61 GAE dearer [pcc:unverified]. Two jumps fail as predicted: 3.25x and 3.46x, with the walk reaching its cap below rank 2 in 94 % and 98 % of runs [pcc:unverified]. Sixteen jumps is at the optimum of this knob at these sizes; the only direction left is more jumps at larger k, where the set-up vanishes (measured in `research/curve25519_tw_scaling_20261108`). Rho-neg remains ahead: the 16-jump arm is 1.36x [1.23, 1.50] rho-neg at k = 20 and 1.34x [1.22, 1.48] at k = 24 at w = 0 on these fresh seeds [pcc:unverified].

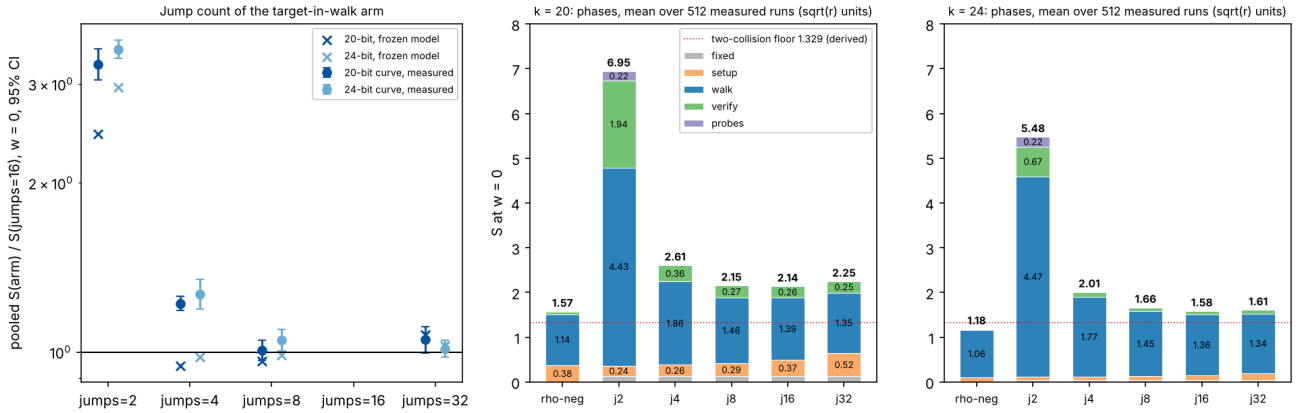


Figure 1: Jump count: pooled ratios and phases

1. Design (as frozen)

8 sessions per size, `targets_per_curve = 1`, 64 measured rounds plus 1 warm-up per arm, `order = alternate`; arms `rho-neg`, `twcj-j16` (baseline), `twcj-j2`, `twcj-j4` (primary), `twcj-j8`, `twcj-j32`, all pipeline arms on `prime-projected-abscissa:size=1,scan_cap=32` with `targets=stride-generator` and `wwalk:x=1.5,g=1,grow=2,tw=1,cj=1,jumps=J,seed=run`. Audit: `ecbench verify --replay 1` per session. Pooling: `scripts/pooled_compare.py` (unchanged copy of the earlier study's, warm-up records skipped), ratio of per-session sums, linearised t interval, df = 7. Runtime: 90 s of sessions at k = 20 and 2 min at k = 24 (`data/run-*.log`).

2. Results (measured)

All 16 sessions completed with 390 of 390 records verified each, and every audit reproduced its replay with 0 problems (`data/verify-k*-s*.txt`) [pcc:unverified]. Pooled figures use the 3,072 measured records per size (`data/pooled_measured.json`).

Mean S per arm (committed band, then with table inserts also priced):

k	arm	S w=0	S w=0.1	S w=1	w=0.1, inserts priced	w=1, inserts priced
20	rho-neg	1.573	1.686	2.711	1.686	2.711
20	twcj-j16	2.136	2.140	2.181	2.279	3.567
20	twcj-j2	6.946	6.976	7.246	7.419	11.676
20	twcj-j4	2.606	2.611	2.658	2.797	4.518
20	twcj-j8	2.153	2.158	2.199	2.304	3.662
20	twcj-j32	2.245	2.250	2.290	2.385	3.642
24	rho-neg	1.179	1.285	2.240	1.285	2.240

24	twcj-j16	1.583	1.584	1.595	1.720	2.950
24	twcj-j2	5.475	5.498	5.705	5.946	10.180
24	twcj-j4	2.008	2.010	2.022	2.187	3.797
24	twcj-j8	1.663	1.664	1.674	1.809	3.124
24	twcj-j32	1.606	1.607	1.617	1.740	2.952

Pooled ratios with 95 % intervals over 8 single-target sessions (committed band):

k	ratio	w = 0	w = 0.1	w = 1
20	j4 / j16	**1.220** [1.186, 1.256]	1.220 [1.186, 1.255]	1.219 [1.186, 1.253]
20	j8 / j16	1.008 [0.967, 1.051]	1.008 [0.967, 1.051]	1.008 [0.968, 1.050]
20	j32 / j16	1.051 [0.996, 1.110]	1.051 [0.996, 1.110]	1.050 [0.996, 1.108]
20	j2 / j16	3.252 [3.055, 3.463]	3.259 [3.061, 3.471]	3.323 [3.120, 3.539]
20	j16 / rho-neg	1.358 [1.232, 1.497]	1.269 [1.149, 1.401]	0.804 [0.722, 0.896] *
20	j4 / rho-neg	1.657 [1.522, 1.805]	1.548 [1.419, 1.690]	0.981 [0.890, 1.081] *
24	j4 / j16	**1.268** [1.193, 1.348]	1.268 [1.193, 1.348]	1.268 [1.192, 1.349]
24	j8 / j16	1.050 [1.005, 1.097]	1.050 [1.005, 1.097]	1.050 [1.005, 1.097]
24	j32 / j16	1.014 [0.978, 1.051]	1.014 [0.978, 1.051]	1.014 [0.978, 1.051]
24	j2 / j16	3.458 [3.331, 3.590]	3.470 [3.342, 3.603]	3.578 [3.445, 3.717]
24	j16 / rho-neg	1.343 [1.219, 1.479]	1.233 [1.118, 1.359]	0.712 [0.643, 0.787] *
24	j4 / rho-neg	1.703 [1.566, 1.852]	1.564 [1.437, 1.701]	0.903 [0.827, 0.985] *

* The committed band prices rho-neg's per-step canonicalisation and not the walk arms' per-step table inserts (the asymmetry found in `research/curve25519_f1_targetwalk_20261108` section 5). With `--price-table-work` (PR #3700's opt-in reading; `data/band-tw-*.txt`, and `ratios_table_work_priced` in `data/pooled_measured.json`) j16 / rho-neg is 1.358 / 1.351 / 1.316 at k = 20 and 1.343 / 1.338 / 1.317 at k = 24 at w = 0 / 0.1 / 1, and j4 / rho-neg is 1.657 / 1.659 / 1.667 and 1.703 / 1.702 / 1.695, so rho-neg is ahead at every weight under that reading [pcc:unverified]. ecbench's own committed band verdict for rho-neg against every twcj arm is "unresolved inside the band" in every session (`data/band-k*.txt`), so no rank between them is admitted under the committed reading [pcc:unverified].

Phases and walk diagnostics (mean over 512 measured runs per arm, units of sqrt(r))

k	arm	fixed	set-up	walk	collision verify	probes	total	walk steps / sqrt(r) (sd)	collisions per run	cap reached below rank 2
20	twcj-j2	0.125	0.237	4.430	1.936	0.218	6.946	4.43 (0.36)	15.19	93.6 %
20	twcj-j4	0.125	0.256	1.860	0.365	0.000	2.606	1.86 (0.66)	2.93	0.2 %
20	twcj-j8	0.125	0.293	1.464	0.271	0.000	2.153	1.46 (0.52)	2.19	0
20	twcj-j16	0.125	0.368	1.386	0.256	0.000	2.136	1.39 (0.49)	2.08	0
20	twcj-j32	0.125	0.518	1.352	0.250	0.000	2.245	1.35 (0.49)	2.03	0
24	twcj-j2	0.040	0.073	4.474	0.673	0.216	5.475	4.47 (0.21)	17.00	98.1 %
24	twcj-j4	0.040	0.078	1.775	0.114	0.001	2.008	1.77 (0.70)	2.97	0.4 %
24	twcj-j8	0.040	0.089	1.450	0.085	0.000	1.663	1.45 (0.51)	2.19	0
24	twcj-j16	0.040	0.110	1.355	0.079	0.000	1.583	1.36 (0.47)	2.05	0
24	twcj-j32	0.040	0.152	1.335	0.079	0.000	1.606	1.34 (0.47)	2.03	0

The set-up follows the frozen model to within 5 % (measured 85.8 / 92.6 / 106.2 / 133.4 / 187.8 GAE for 2 / 4 / 8 / 16 / 32 jumps at k = 20, about 3.5 GAE per jump plus 80 fixed) [pcc:unverified]. The walk does not: every reduction in jumps lengthens it, and every collision beyond the second is a dependent row [pcc:unverified].

3. Predictions (PROTOCOL.md section 5)

id	prediction	outcome	status

P1 (primary)	k=20 j4/j16 upper end < 1 at w = 0, 0.1, 1	1.256, 1.255, 1.253	**failed** [pcc:unverified]
P1b	k=20 j4/j16 point in [0.92, 0.98]	1.220	**failed** [pcc:unverified]
P2	k=20 j8/j16 point in [0.93, 0.99]	1.008	**failed** [pcc:unverified]
P3	j4, j8 walk steps within 3 % of j16, both sizes	j4 +34 % / +31 %, j8 +5.6 % / +7.0 %	**failed** [pcc:unverified]
P4	k=20 j32/j16 lower end > 1	0.996	**failed** (by 0.004) [pcc:unverified]
P5	j2/j16 lower end > 1.5 and cap below rank 2 in >= 90 % of runs, both sizes	3.055 / 3.331; 93.6 % / 98.1 %	**held** [pcc:unverified]
P6	k=24 j4/j16 point in [0.96, 1.00]	1.268	**failed** [pcc:unverified]
P7	j4/rho-neg lower end > 1 at w = 0 and, inserts priced, at w = 0, 0.1, 1, both sizes	1.522, 1.520, 1.512; 1.566, 1.564, 1.552	**held** [pcc:unverified]
P7b	k=20 j4/rho-neg point in [1.25, 1.40]	1.657	**failed** [pcc:unverified]

Two of nine held. The failed predictions stay failed; nothing was re-scored.

4. Why the model failed (exploratory, derived from the counters, not frozen)

The model assumed that with m >= 4 jumps the collision count stays n^2/r and that every collision is an independent row. The counters show two effects it missed [pcc:unverified]:

- **Dependent re-collisions.** After a collision E_t = E_s the walk continues from a point it has already stored, and because the jump index is drawn independently of the point, the next steps from E_t repeat the stored path from E_s whenever the two short jump sequences have the same count vector. With m jumps that happens with probability about 1/m per step for a one-step repeat (and more for longer matches), and every such re-collision is the earlier row again, so rank 2 needs a third, fourth, … collision. Measured collisions per run: 2.93 (m = 4), 2.19 (m = 8), 2.08 (m = 16), 2.03 (m = 32), against the two that rank 2 needs [pcc:unverified].
- **A slower approach to uniformity.** Even the first independent row arrives later with fewer jumps (the per-collision rate falls below 2/r because the difference vectors of nearby positions repeat); the walk-step means 1.86 / 1.46 / 1.39 / 1.35 sqrt(r) for 4 / 8 / 16 / 32 jumps at k = 20 fall monotonically toward the 1.329 floor [pcc:unverified].

Both effects favour more jumps. At these sizes each jump costs about 3.5 GAE of set-up, which cancels the gain already at 32 jumps (walk 0.034 sqrt(r) shorter, set-up 0.150 sqrt(r) dearer at k = 20). At k = 28 and 32 the set-up is 0.01 sqrt(r) or less, which is the follow-on measured in `research/curve25519_tw_scaling_20261108` [pcc:unverified].

5. Distance to matched rho (rho-neg, same sessions)

k	arm	w = 0	w = 0.1 (committed / inserts priced)	w = 1 (committed / inserts priced)
20	twcj-j16 (current best)	1.36x [1.23, 1.50]	1.27x / 1.35x	0.80x (artifact) / 1.32x
20	twcj-j32	1.43x	1.33x / 1.41x	0.85x (artifact) / 1.34x
24	twcj-j16	1.34x [1.22, 1.48]	1.23x / 1.34x	0.71x (artifact) / 1.32x
24	twcj-j32	1.36x	1.25x / 1.35x	0.72x (artifact) / 1.32x

No arm of this study moves the distance remaining; the 16-jump arm stays the best, at 1.34x to 1.36x rho-neg at w = 0, with the earlier studies' 1.40x and 1.41x inside these intervals [pcc:unverified]. Rho-neg's own set-up (0.376 sqrt(r) at k = 20, 0.098 at k = 24) is part of the reference, as in the earlier studies.

6. Scope and limits

- The F = 1 target-in-walk arm is one unknown away from rho: with P1 := G it is a random walk on [c]G + [b]Q whose first collision gives log Q (rho with a stored walk). It does no decomposition work and is not decomposition-based index calculus; no leaderboard or `benchmarks/README.md` row is proposed, and none could be added without the validation gate (no VALIDATION.json, no replay certificate, no validator report) [pcc:unverified].
- Single target only; the walk is built from the target.
- Local L0 counted units on a shared host; deterministic and replayed; no wall-clock claim.
- No A/A arm (ecbench seeds runs by workload and round; an A/A arm replicates rho-neg exactly, as the earlier studies found). Noise is carried by the between-session t intervals.
- research-kb and ecdb were not reachable from this machine; no novelty claim is made.

Files

`PROTOCOL.md`, `specs/`, `data/frozen.sha256`, `data/frozen-at.txt`, `scripts/` (`model.py`, `make_specs.py`, `run_sessions.sh`, `pooled_compare.py`, `analyse.py`, `make_figure.py`, `render_pdf.py`), `sessions/k*-s*/{session,spec,plan,host}.json` (the

records.jsonl files are in data/sessions-k{20,24}-records.tar.gz, hashed in the benchmark records), data/ (model, plans, run logs, both band readings, verify reports, pooled_measured.json, binary hashes), figures/jumps.{svg,png}, REPORT.pdf. Benchmark records: benchmarks/records/20261108T2045Z-curve25519-tw-jumps-k20.json, benchmarks/records/20261108T2047Z-curve25519-tw-jumps-k24.json; figure and report copies in benchmarks/20261108-curve25519-tw-jumps/.